

Introduction To Geometric Construction

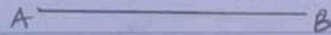
Introduction To Geometrical Construction

In this chapter we would learn step by step method to construct angles with just pencil, compass and ruler.

i) Construction of angle 90°

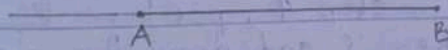
step 1

Draw any arbitrary line, that is, any line of any convenient length as shown below



step 2

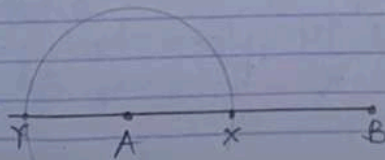
Let's say we wish to draw a perpendicular line to the point A. We will extend the line from A a little bit to the left.



NOTE: A perpendicular line is a line that angle 90° to the given line.

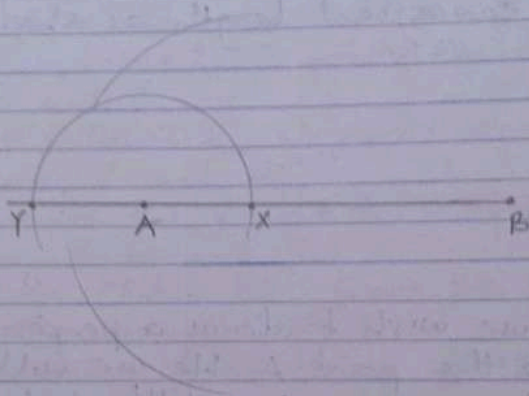
step 3

With point A as the center, draw an arc of any convenient radius to touch line AB and the extended line as shown below



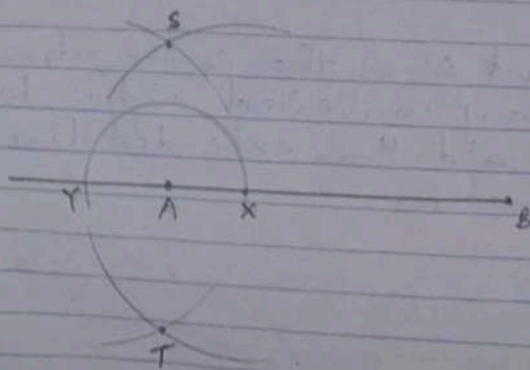
Step 4
He labelled the arc points on the line as points X and Y.

With point X as the center, and the radius greater than XA , use the compass to draw two arcs above the line AB as shown below.

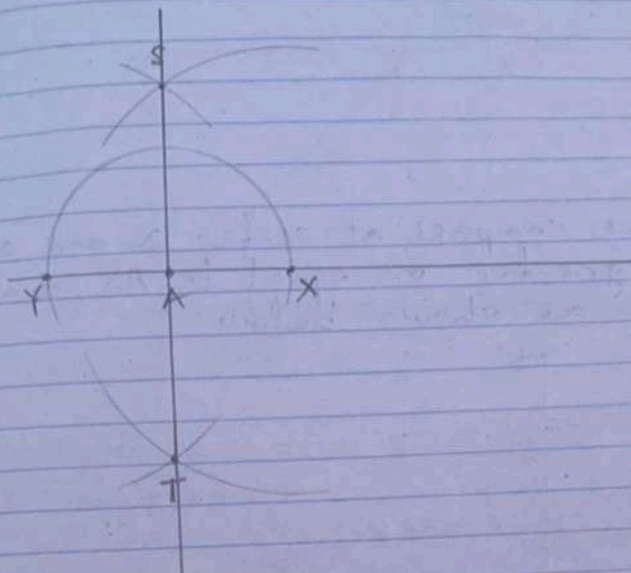


Step 5

With the same radius used to draw arcs above line AB in step 4, position the compass at center Y and draw two arcs above AB to join the arcs drawn in step 4 above.



step 6
Finally join the point S and T through point A.



The angle at A, that is $\angle SAX = 90^\circ$
and $\angle SAY = 90^\circ$

Also note that the above is also the procedure for perpendicularly bisecting a straight line at a given point.

ii) Construction of angle 45°

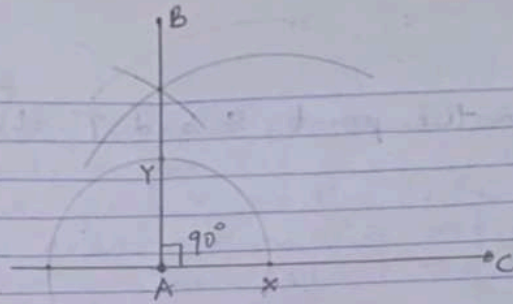
To construct angle 45° , we would bisect angle 90° .

Bisection of an angle means to divide the angle into two equal parts.

So bisecting 90° , means dividing it into angle 45° .

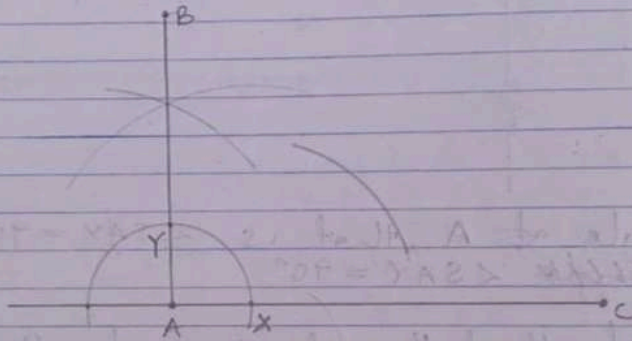
step 1

Construct angle 90° by drawing a line AB perpendicular to the line AC



Step 2

With the compass at center X and a radius greater or equal to AX, draw an arc as shown below



Step 3

With the same radius used in step 2 above and positioning the compass at Y as the center, draw an arc to cut the arc drawn in step 2 above

